MAINTENANCE SCHEDULE

NOTE: BEFORE WORKING, CLEANING OR MAINTAINING THE EQUIPMENT, IT IS HIGHLY IMPORTANT TO LOCK OUT ALL OF THE ENERGY SOURCES ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

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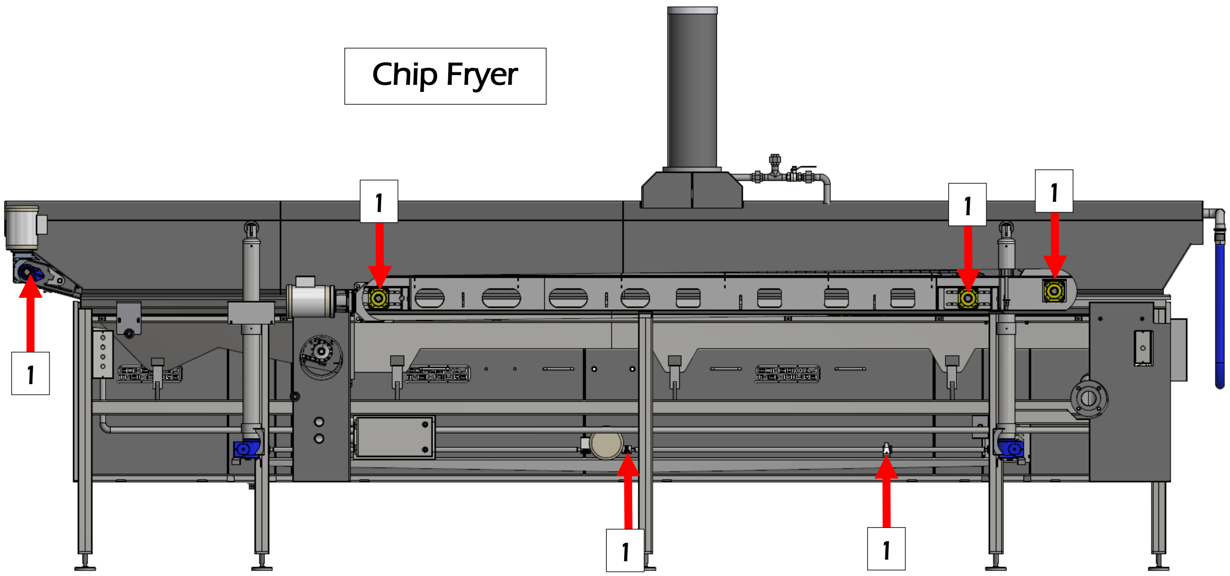
LUBRICATION SCHEDULE



Item	Part	LE #	Grease/Oil Description	When
1	Bearing*	MH027-02	General purpose grease, food grade	Every 168 hours of operation
2	Bearing*	MH027-05	Hi-Temp Non Graphite Grease	NA
3	Bearing	MH027-01	Hi-Temp w/Graphite Grease	NA
4	Chain (oven)	MH027-15	Chain Oil (for Auto Chain Oiler)	NA
5	Chain (oven)	MH027-03	Chain Oil (manually applied)	NA
6	Chain (proofer)	MH027-31	Chain Oil (for Auto Chain Oiler)	NA
7	Chain (proofer)	MH027-31	Chain Oil (manually applied)	NA
8	Oil Tank	MH027-06	Hydraulic Oil	NA

- * Grease bearings through zerk fittings provided on bearings. DO NOT force excess amount of grease through the seals. If the bearings become over pressured by too many pumps on a grease gun, the main seals may become damaged or extruded from the bearings. This will cause for the bearing not to hold any grease. Bearing failure will occur. Clean the outsides thoroughly after lubing.
- * Motors and speed reducers, etc. are factory set.

Cleaning Chemicals

The main chemical for cleaning fryers is Caustic Soda (Sodium Hydroxide – NaOH) in either a liquid or flake form.

It is most often used in conjunction with other constituents such as, wetting agents, suspension agents, detergents, emulsifiers, sequestrates, oxidizers, and corrosion inhibitors.

If these additional constituents are used in a CIP system, then a low foaming type should be selected.

Safety Precautions and Personal Protection Equipment

Suitable time should be allowed for thorough cleaning of industrial fryers and care taken for safety of personnel with use of adequate Personal Protection Equipment (PPE).

WARNING

The detergent is highly caustic and **will cause serious injuries** if due care is not taken.

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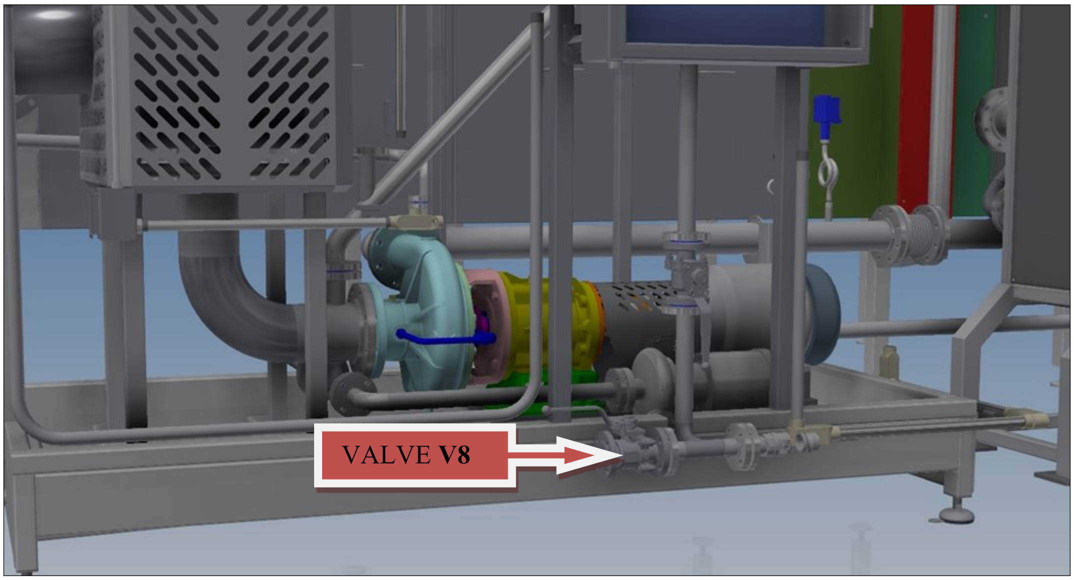
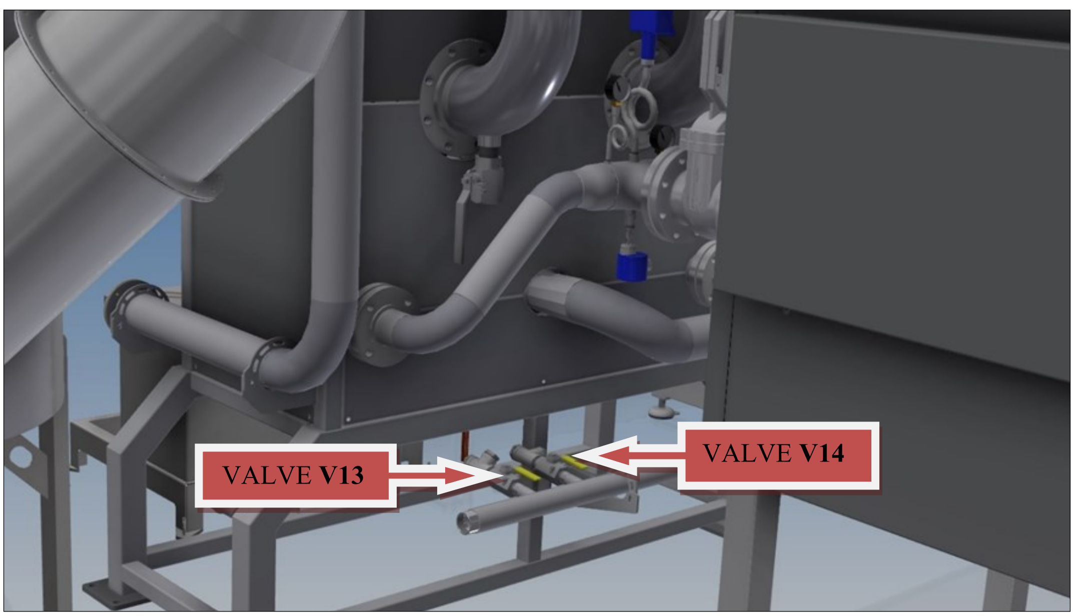
SYSTEM DRAINING PRIOR TO CIP CLEANING CYCLE

- To drain the cooking oil circuit properly, open valve **V8, V13** and **V14** until all the oil is drained.





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FRYER CAUSTIC CHEMICAL CLEANING

Caution: Wear the appropriate safety gear when mixing caustic solution. Wear rubber overalls, mask, safety goggles and gloves.

Start the Thermofluid boilers, set Thermofluid temperature to 275 degrees Fahrenheit. Start Thermofluid boilers and circulation pump on boilers control panel.

1. Connect the hood spray bar CIP hose to the connection on discharge end of hood. Make sure cam locks are engaged and the valve supplying solution to the sprayers is open.
2. Ensure the system has been drained of all residual cooking oil. Open drain valves V8, V13 and V14.
3. Raise the kettle hood and remove any settled debris from the kettle bed and conveyor frame.
4. Remove covers from filter drum box, remove the scrap conveyor. Clean off any debris. Replace the conveyor and filter drum box covers once cleaned up.
5. Set all valves according to valve position chart for “CIP”. Fill system with fresh water. Water volume is approximately 290 gallons
6. Lower the kettle hood all the way down until it is stopped by the low limit switch.
7. Start the fryer pump, conveyors, drums and oil filter system.
8. Slowly add the amount of caustic soda recommended by your Chemical Company above the oil outlet nozzle on the infeed side of the fryer kettle. (Approximately 60lbs Caustic Soda Powder for a full system wash)
9. Set water temperature to 205F.
10. Cleaning time will depend on frequency of cleaning. If cleaned once weekly an approximate boil time would be between 3 and 6 hours.
11. At the end of this cleaning cycle, Stop the conveyors, drum, oil filter system and circulating pump.
12. Raise the hood. Scrub the side walls of the kettle, conveyor frame and drums to remove any oil residue or debris.
13. Drain the caustic wash solution from the fryer by setting the valves in the “Draining” position on the valve chart.
14. Leaving the drain valves open, rinse the fryer conveyor, underside of hood and internal components with fresh water. Rinse the filter drum box, drum and scrap conveyor with fresh water.
15. After rinsing, allow the system to drain. When empty, set the valves to the “CIP” setting on the valve position chart. Fill the system with warm water.
16. Run the circulating pump, paddles, conveyor and filter system for 30 minutes with fresh water.
17. Shut down the Thermofluid boilers at this time. No heat from them is needed in next steps.

FRYER CAUSTIC CHEMICAL CLEANING (Continued)

Caution: Wear the appropriate safety gear when mixing caustic solution. Wear rubber overalls, mask, safety goggles and gloves.

18. Raise the hood and rinse the fryer conveyor, frame and drums between water fill cycles. Rinse the filter drum system and scrap conveyor. Lower the hood after rinsing to continue cycle.

19. Continue the drain and fill cycle until the PH of the water coming out of the fryer matches the PH of the fresh plant water. Use new, uncontaminated PH strips to test the water.

20. When the water PH exiting the fryer kettle matches plant supply, completely drain the system and give a final rinse to the underside of kettle hood, conveyors, drums, filter drum system and scrap conveyor

21. Remove the flexible hose from the cam lock and drain the solution from the lines. Allow the flexible line to drain completely.

22. Oil mist eliminator filters may be left in during cleaning cycle or removed and cleaned separately. The backside of the filters may not be thoroughly cleaned during cleaning cycle, as there are no spray jets on the backside of the filters.

23. Set all valves according to draining column on chart . Ensure all low point drain valves are open. Ensure all drain plugs have been removed. Make sure all water has been drained.

24. Verify that all valves are returned to their proper position and that all plugs are replaced before filling the fryer with fresh oil.

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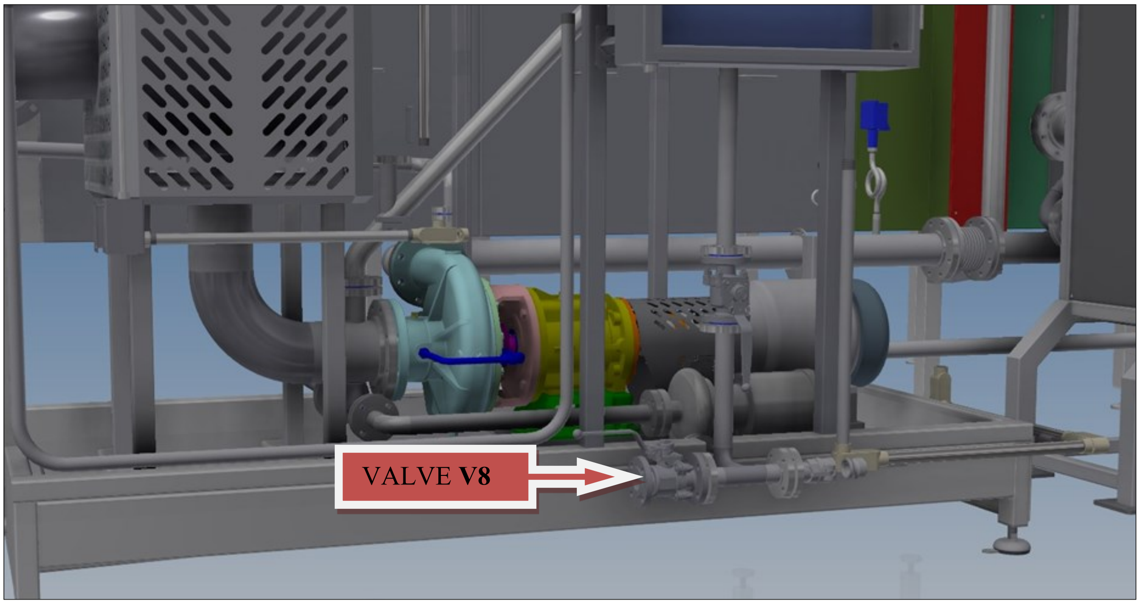
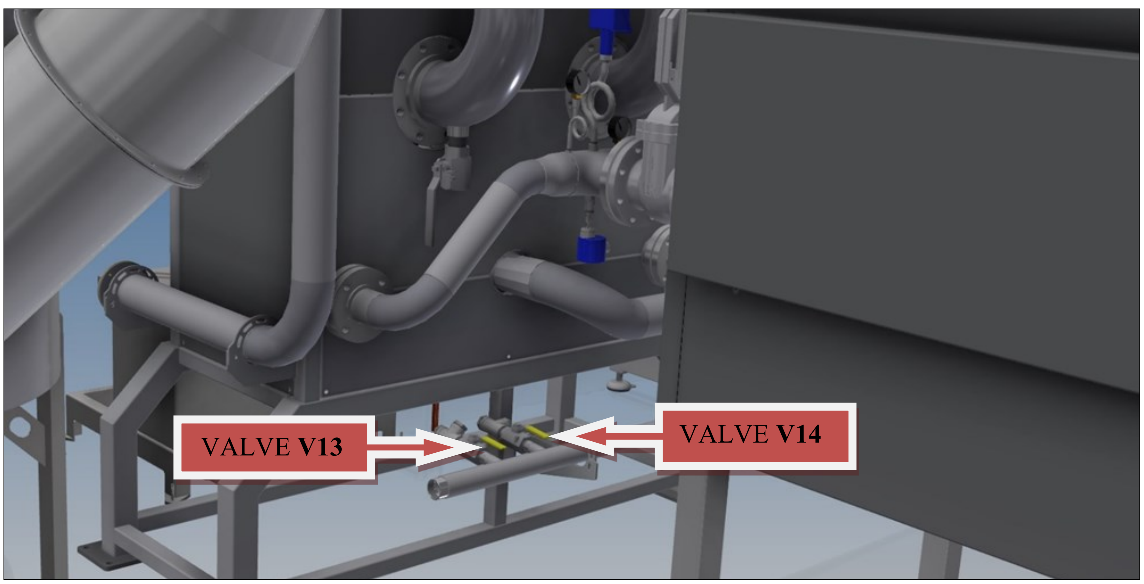
COOKING OIL DRAINING VALVES BE SURE TO OPEN ALL COOKING OIL DRAINING VALVES BEFORE RE-FILLING THE SYSTEM WITH OIL





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CLEANING PLC SCREEN

1. Gently wipe the dust using a dry, lint-free cloth, like a microfiber cloth. Remove any tags that may be on the cloth to avoid scratching the display with tags.
2. Lightly damp a microfiber cloth with water and gently wipe the display with as little pressure as possible.
3. If you prefer, you can use a cleanser made specifically for cleaning LCD displays, but **Do Not** spray directly onto the PLC screen. Spray a small amount of the cleaner on a microfiber cloth first, to avoid getting any cleaning solution inside the display.
4. Do not use the following product agents:

* Ammonia
* Ethyl alcohol
* Acetone
* Ethyl acid
* Methyl chloride

Those cleaning agents will damage the finish and any special coatings that may be on the display.

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

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CLEANING CHIP FRYER CRUMB CONVEYOR

1. If this is a full CIP water and solution cleaning, ensure the fryer kettle and filter box have been pumped empty and drained of all cooking oil. NEVER attempt to remove Crumb Conveyor when there is hot oil circulating through the system. Severe injury can occur.
2. Do not remove the conveyor or components if the sheet metal and surrounding areas are still hot from the production run. Severe burns can happen by touching the hot metal. Wait until the surfaces have cooled down.
3. Turn Off and lock out the Crumb Conveyor motor disconnect. *See Fig. A*
4. After locking out the Crumb Conveyor disconnect, remove the motor plug using a counter clockwise twist then pull motion. Be careful not to bend the plug tabs. Ensure the motor cord female box cover is closed or covered so no water can enter the receptacle. *See Fig. B*

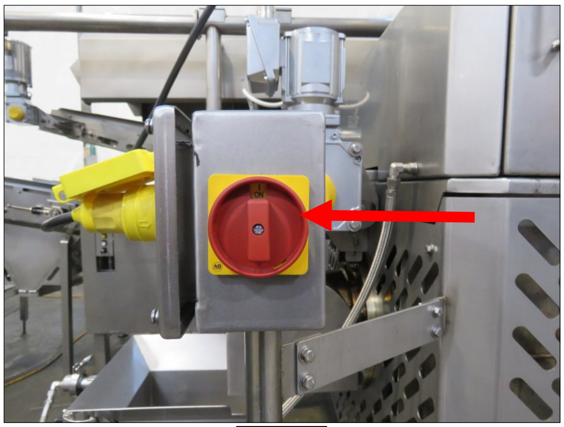

Fig. A

Fig. B

5. For safety, use safety glasses or face shield and gloves when cleaning the Crumb Conveyor and the related components.
6. Remove the two top filter box covers. *See Fig. C*
7. Remove the Crumb Conveyor mesh filter by loosening the retaining bolt under the front center of the filter drip pan. Slide the mesh filter straight out. *See Fig. D*

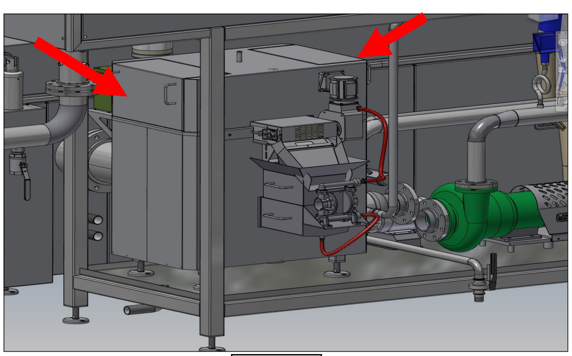

Fig. C

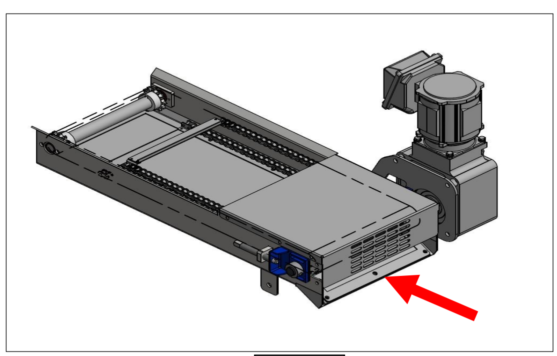

Fig. D

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CLEANING CHIP FRYER CRUMB CONVEYOR (continued)

8. Brush off the Crumb Conveyor mesh filter with a soft brush to remove residual crumbs.

9. Rinse the mesh filter with clean water, apply cleaning solution if the filter has heavy build up. Always rinse the filter with fresh water after any cleaning solutions have been applied.

10. Remove the mesh filter drain oil pump return line from the top cover of the Crumb Conveyor enclosure. *See Fig. E*

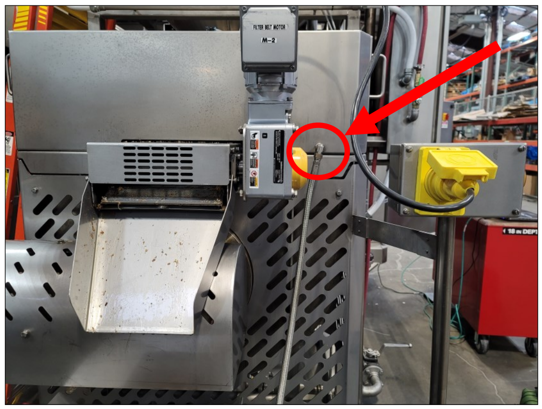

Fig. E

11. Remove the filter drum blow off supply hose by pushing in on the outer perimeter of where the hose inserts then pull straight out. *See Fig. F*

12. Remove the four mounting bolts for the enclosure cover. There are eight bolts, one in each corner and two in each center area. Be careful not to drop the bolts or washers in to the filter drum box. *See Fig. F*

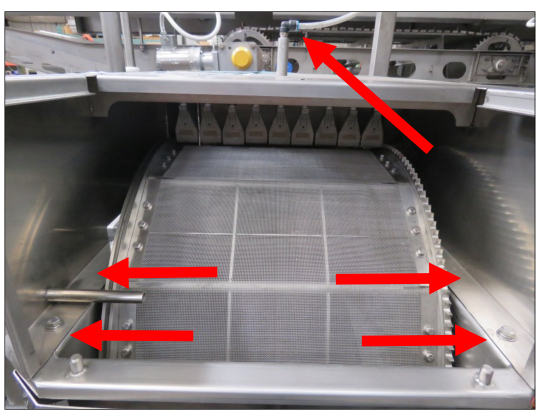

Fig. F

13. Lift the cover straight up to remove. Rinse the cover with water and a cleaning solution.

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CLEANING CHIP FRYER CRUMB CONVEYOR (continued)

14. Remove the Crumb Conveyor retaining shaft from the underside of the frame. There is one shaft going left to right. *See Fig. G*
15. Remove the Crumb Conveyor crumb chute from the conveyor frame by lifting up off of the corner shoulder bolts. *See Fig. G*

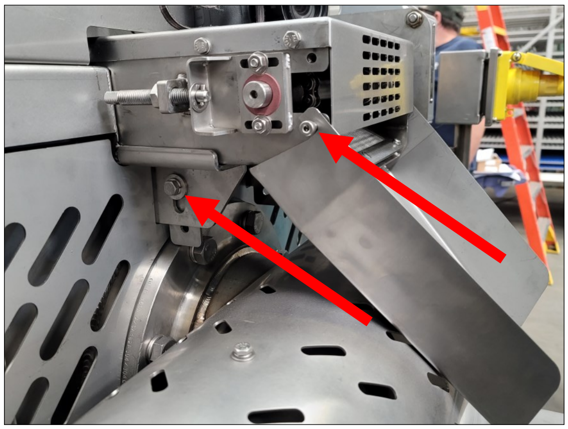

Fig. G

16. Remove the Crumb Conveyor by sliding straight out of the filter enclosure on the mount channels.
17. Remove all debris and crumbs from the Crumb Conveyor mounting channels, drum parts, inside of drum, etc. Rinse with cleaning solution and fresh Hot water.
18. Remove all crumbs and debris from the Crumb Conveyor, chains and pan. Rinse clean with cleaning solution and fresh Hot water. Blow off excess water with debris using clean compressed air.
19. When all parts are clean, reinstall all the components.
20. Slide the Crumb Conveyor into the filter enclosure.
21. Install the Crumb Conveyor retaining bolts into the brackets under the pan.
22. Install the Crumb Conveyor crumb chute by setting tabs onto shoulder bolts.
23. Install the filter box enclosure cover, install the mounting bolts in the corners and center areas. Be careful not to drop the bolts into the filter box.
24. Install the filter drum blow off air hose into the fitting on top of the filter box cover.
25. Install the oil return hose back onto the front of the enclosure cover.
26. Install the filter mesh frame, install the retaining bolt under the crumb conveyor pan.
27. Install the filter box top side covers.
28. Connect the Crumb conveyor motor plug to the receptacle.
29. Remove lock out and then Turn on the Crumb conveyor motor disconnect switch.

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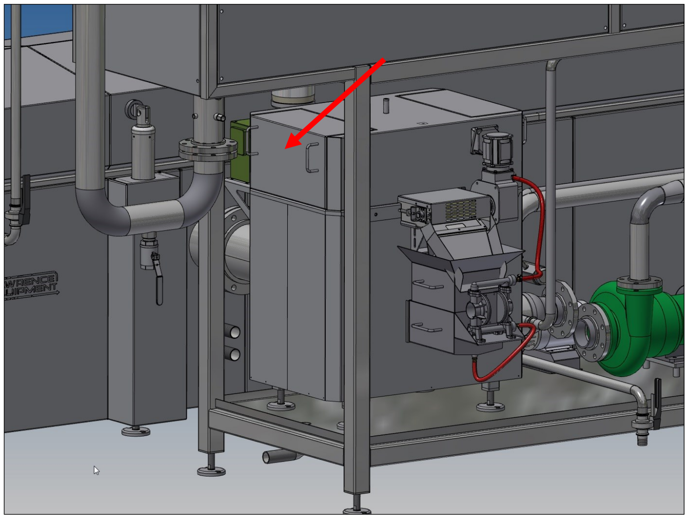
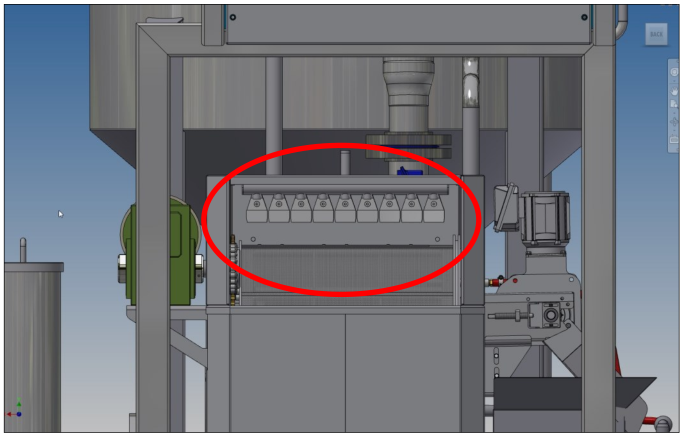
CLEANING FRYER DRUM FILTER ASSEMBLY

1. Remove the oil from the fryer.
2. Make sure the machine has cooled down to room temperature.
3. Use safety when cleaning machine, it is recommended to use gloves and safety glasses.
4. Remove fryer filter cover.



5. Manually clean air nozzles using a cleaning solution.
6. Wipe clean with clean cloth.



7. Reinstall the filter cover.
8. The drum filter should be cleaned when running cleaning cycle. Using caustic cleaning solutions recommended by caustic manufacturer.
9. Make sure air nozzles and filter drum are cleaned weekly or as needed.

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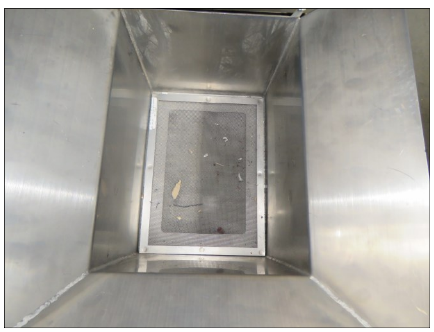
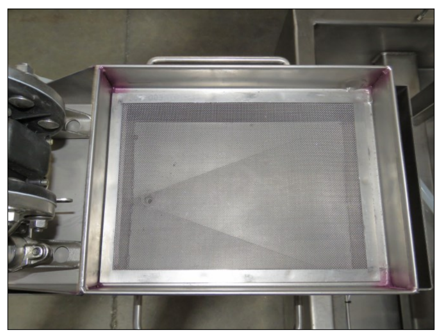
DAILY CLEANING

1. Empty Crumb conveyor filter daily or as needed. Wash with antibacterial soap then rinse with hot water.

2. Remove oil strainer filters daily or as needed. Wash with antibacterial soap then rinse with hot water.





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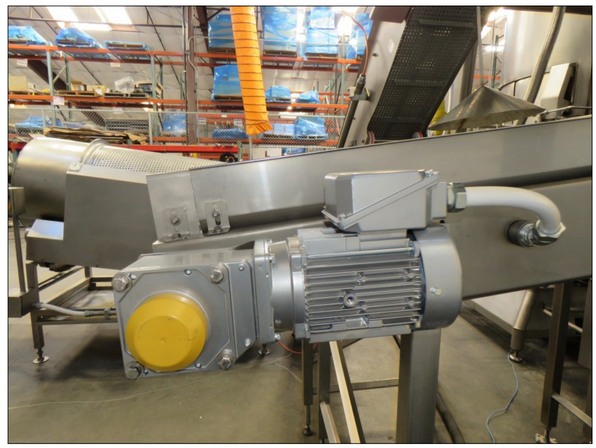
CLEANING INFEED BELTS

Daily cleaning-

1. Make sure to lockout and tagout out machine then Disconnect the Infeed power plug.
2. Unlock the four swivel wheels. Row conveyor out to cleaning area.
3. Loosen any debris or product deposits from wire belt conveyor with a cloth or soft nylon brush.
4. Wash down conveyor with a detergent solution. Make sure to cover all motors switches before washing conveyor.



Weekly cleaning-

1. Remove panel and guards. Wash down the removable panels and guards with a detergent solution. Rinse with Fresh water, wipe down with clean cloth to dry.
2. Loosen product deposits and debris from wire conveyor and brush away. Wash with detergent solution then rinse with fresh water. Wipe down clean cloth to dry.
3. Remove product deposits from the Conveyor belt surface with a soft nylon brush. Make sure compacted debris is removed from the sprockets, idler wheels and support rails. Pre-rinse the belt and support system with warm water. Apply an appropriate foaming detergent mixture to the belt and support system. Rinse the belt and support system with clean water. After the rinse, inspect the belt and support system components to ensure it is free of any residue. Verify that all cleaning chemical is removed from the conveyor belt, sprockets, idlers, and support rails. Its recommended that pH testing be used as an aid in determining that the belt is free of the detergent. Run the conveyor belt slowly to help dry it and its supports, and remove any pooled water from the floor.
4. Degrease drive chain, motor, etc. with solvent if needed. Wipe with clean cloth dampened in clean water.
5. Scrub down interior framework, ledges, etc., with detergent solution then rinse with clean water.
6. Replace all cleaned panels, guards, etc. when thoroughly dried. Make sure all standing or pooled water is dried by using clean cloth or clean compressed air.